

go up or down, and performing management functions such as those we'll study in Chapter 9—operate at the millisecond or second timescale. These **router control plane** functions are usually implemented in software and execute on the routing processor (typically a traditional CPU).

Before delving into the details of a router's control and data plane, let's return to our analogy of Section 4.1.1, where packet forwarding was compared to cars entering and leaving an interchange. Let's suppose that the interchange is a roundabout, and that before a car enters the roundabout, a bit of processing is required—the car stops at an entry station and indicates its final destination (not at the local roundabout, but the ultimate destination of its journey). An attendant at the entry station looks up the final destination, determines the roundabout exit that leads to that final destination, and tells the driver which roundabout exit to take. The car enters the roundabout (which may be filled with other cars entering from other input roads and heading to other roundabout exits) and eventually leaves at the prescribed roundabout exit ramp, where it may encounter other cars leaving the roundabout at that exit.

We can recognize the principal router components in Figure 4.6 in this analogy—the entry road and entry station correspond to the input port (with a lookup function to determine the local outgoing port); the roundabout corresponds to the switch fabric; and the roundabout exit road corresponds to the output port. With this analogy, it's instructive to consider where bottlenecks might occur. What happens if cars arrive blazingly fast (for example, the roundabout is in Germany or Italy!) but the station attendant is slow? How fast must the attendant work to ensure there's no backup on an entry road? Even with a blazingly fast attendant, what happens if cars traverse the roundabout slowly—can backups still occur? And what happens if most of the entering cars all want to leave the roundabout at the same exit ramp—can backups occur at the exit ramp or elsewhere? How should the roundabout operate if we want to assign priorities to different cars, or block certain cars from entering the roundabout in the first place? These are all analogous to critical questions faced by router and switch designers.

In the following subsections, we'll look at router functions in more detail. [Iyer 2008, Chao 2001; Chuang 2005; Turner 1988; McKeown 1997a; Partridge 1998] provide a discussion of specific router architectures. For concreteness, the ensuing discussion assumes a datagram network in which forwarding decisions are based on the packet's destination address (rather than a VC number in a virtual-circuit network). However, the concepts and techniques are quite similar for a virtual-circuit network.

4.3.1 Input Processing

A more detailed view of input processing is given in Figure 4.7. As discussed above, the input port's line termination function and link-layer processing implement the physical and link layers for that individual input link. The lookup performed in the input port is central to the router's operation—it is here that the router uses the forwarding table to look up the output port to which an arriving packet will be

CASE HISTORY

CISCO SYSTEMS: DOMINATING THE NETWORK CORE

As of this writing 2012, Cisco employs more than 65,000 people. How did this gorilla of a networking company come to be? It all started in 1984 in the living room of a Silicon Valley apartment.

Len Bosak and his wife Sandy Lerner were working at Stanford University when they had the idea to build and sell Internet routers to research and academic institutions, the primary adopters of the Internet at that time. Sandy Lerner came up with the name Cisco (an abbreviation for San Francisco), and she also designed the company's bridge logo. Corporate headquarters was their living room, and they financed the project with credit cards and moonlighting consulting jobs. At the end of 1986, Cisco's revenues reached \$250,000 a month. At the end of 1987, Cisco succeeded in attracting venture capital—\$2 million from Sequoia Capital in exchange for one-third of the company. Over the next few years, Cisco continued to grow and grab more and more market share. At the same time, relations between Bosak/Lerner and Cisco management became strained. Cisco went public in 1990; in the same year Lerner and Bosak left the company.

Over the years, Cisco has expanded well beyond the router market, selling security, wireless caching, Ethernet switch, datacenter infrastructure, video conferencing, and voice-over IP products and services. However, Cisco is facing increased international competition, including from Huawei, a rapidly growing Chinese network-gear company. Other sources of competition for Cisco in the router and switched Ethernet space include Alcatel-Lucent and Juniper.

forwarded via the switching fabric. The forwarding table is computed and updated by the routing processor, with a shadow copy typically stored at each input port. The forwarding table is copied from the routing processor to the line cards over a separate bus (e.g., a PCI bus) indicated by the dashed line from the routing processor to the input line cards in Figure 4.6. With a shadow copy, forwarding decisions can be made locally, at each input port, without invoking the centralized routing processor on a per-packet basis and thus avoiding a centralized processing bottleneck.

Given the existence of a forwarding table, lookup is conceptually simple—we just search through the forwarding table looking for the longest prefix match, as described

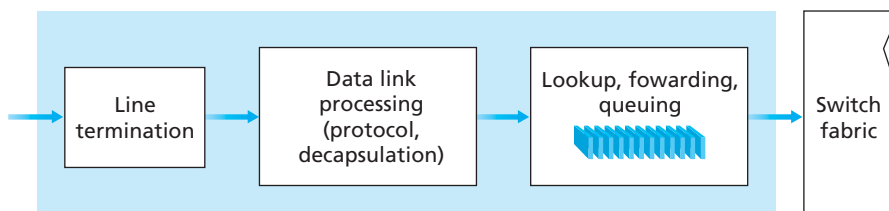


Figure 4.7 ♦ Input port processing

in Section 4.2.2. But at Gigabit transmission rates, this lookup must be performed in nanoseconds (recall our earlier example of a 10 Gbps link and a 64-byte IP datagram). Thus, not only must lookup be performed in hardware, but techniques beyond a simple linear search through a large table are needed; surveys of fast lookup algorithms can be found in [Gupta 2001, Ruiz-Sanchez 2001]. Special attention must also be paid to memory access times, resulting in designs with embedded on-chip DRAM and faster SRAM (used as a DRAM cache) memories. Ternary Content Address Memories (TCAMs) are also often used for lookup. With a TCAM, a 32-bit IP address is presented to the memory, which returns the content of the forwarding table entry for that address in essentially constant time. The Cisco 8500 has a 64K CAM for each input port.

Once a packet's output port has been determined via the lookup, the packet can be sent into the switching fabric. In some designs, a packet may be temporarily blocked from entering the switching fabric if packets from other input ports are currently using the fabric. A blocked packet will be queued at the input port and then scheduled to cross the fabric at a later point in time. We'll take a closer look at the blocking, queuing, and scheduling of packets (at both input ports and output ports) in Section 4.3.4. Although "lookup" is arguably the most important action in input port processing, many other actions must be taken: (1) physical- and link-layer processing must occur, as discussed above; (2) the packet's version number, checksum and time-to-live field—all of which we'll study in Section 4.4.1—must be checked and the latter two fields rewritten; and (3) counters used for network management (such as the number of IP datagrams received) must be updated.

Let's close our discussion of input port processing by noting that the input port steps of looking up an IP address ("match") then sending the packet into the switching fabric ("action") is a specific case of a more general "match plus action" abstraction that is performed in many networked devices, not just routers. In link-layer switches (covered in Chapter 5), link-layer destination addresses are looked up and several actions may be taken in addition to sending the frame into the switching fabric towards the output port. In firewalls (covered in Chapter 8)—devices that filter out selected incoming packets—an incoming packet whose header matches a given criteria (e.g., a combination of source/destination IP addresses and transport-layer port numbers) may be prevented from being forwarded (action). In a network address translator (NAT, covered in Section 4.4), an incoming packet whose transport-layer port number matches a given value will have its port number rewritten before forwarding (action). Thus, the "match plus action" abstraction is both powerful and prevalent in network devices.

4.3.2 Switching

The switching fabric is at the very heart of a router, as it is through this fabric that the packets are actually switched (that is, forwarded) from an input port to an output port. Switching can be accomplished in a number of ways, as shown in Figure 4.8:

- *Switching via memory.* The simplest, earliest routers were traditional computers, with switching between input and output ports being done under direct control of